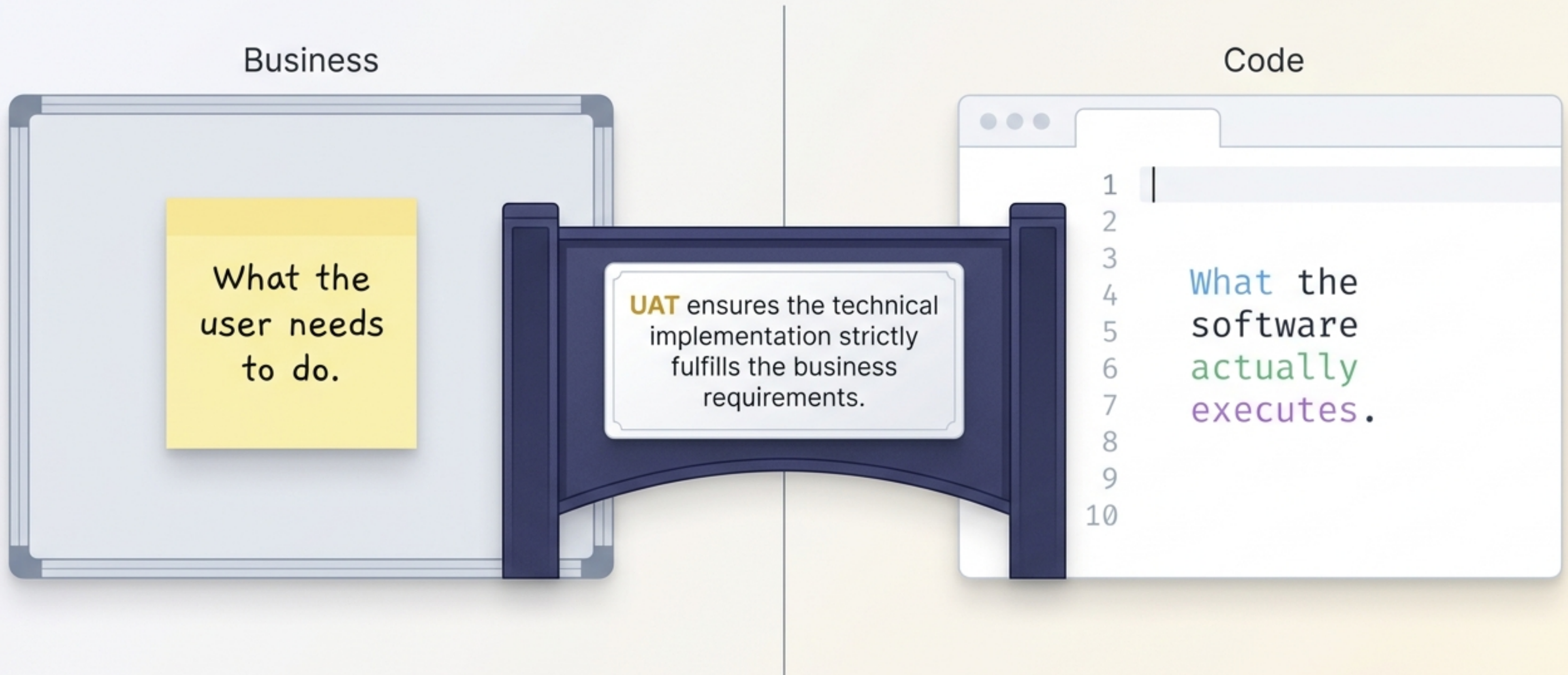


CSE 322 Software Engineering

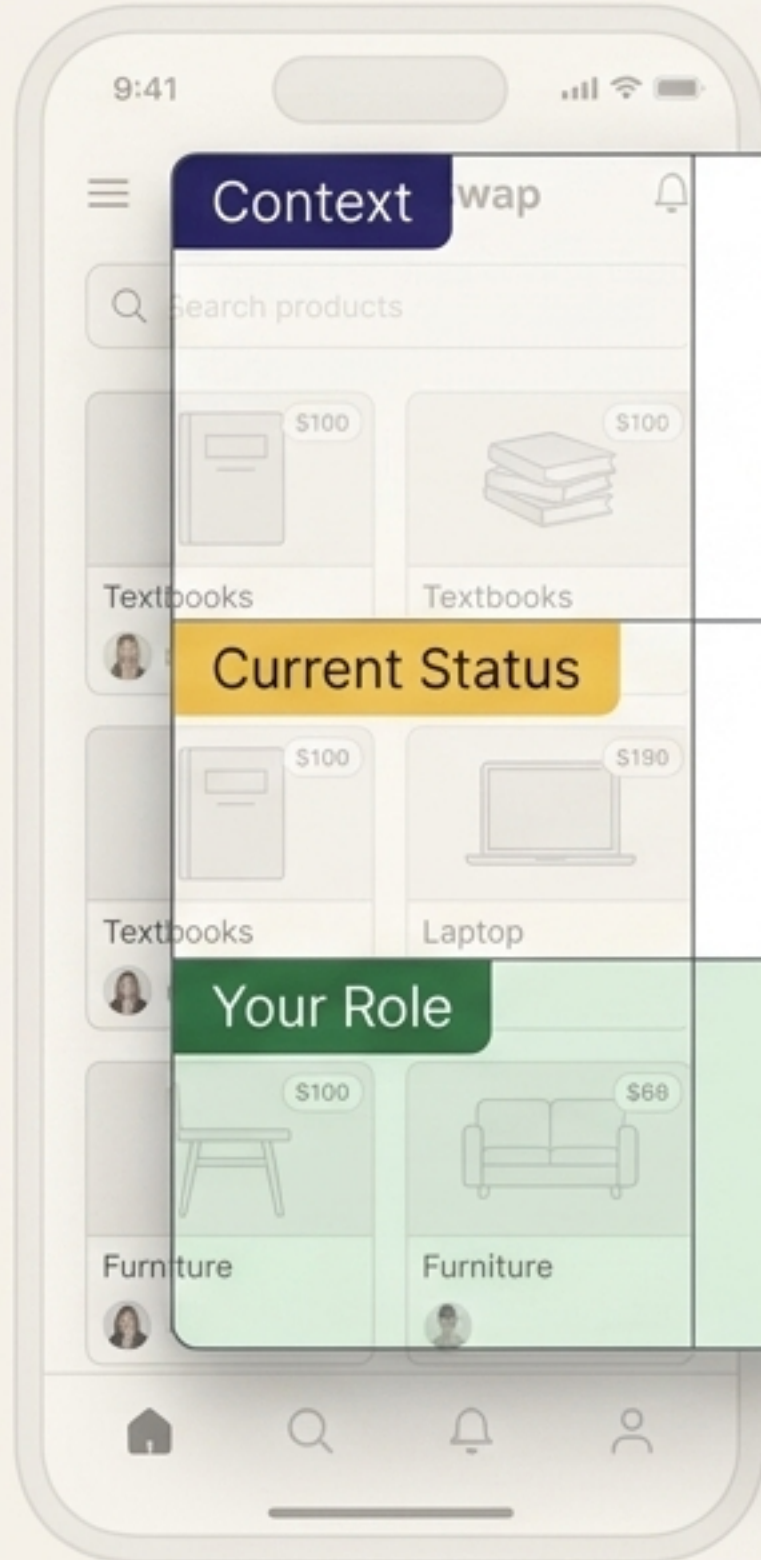
The QA Consultant's Playbook: Mastering Acceptance Testing

Bridging the gap between business requirements and technical implementation through Gherkin and AI.

The Missing Link: User Acceptance Testing (UAT)



Mission Briefing: The 'CampusSwap' Launch



Context

Your university is launching CampusSwap, a peer-to-peer web application for students to buy, sell, or trade textbooks, furniture, and electronics.

Current Status

The development team has just finished the Core Sprint.

Your Role

Quality Assurance (QA) Consultant. You are tasked with verifying that the system is ready for the student body.

Sprint Backlog: Core User Stories

Listing an Item

As a student seller, I want to upload a photo and description of my textbook so that buyers can see its condition.



Searching

As a student buyer, I want to filter items by 'Price' and 'Category' so I can find affordable books quickly.



Secure Communication

As a user, I want to message a seller through an in-app chat without revealing my personal phone number.



The Shared Language: Anatomy of Gherkin Syntax

Gherkin translates human requirements into a structured, industry-standard format understood by humans, developers, and AI.

Given

Setup & State. Establishes the preconditions before the user acts.

When

Trigger & Action. The specific interaction the user performs.

Then

Verification & Outcome. The measurable system response that defines success.

Translating Story to Script

Listing an Item

As a student seller, I want to upload a photo and description of my textbook so that buyers can see its condition.



```
1 Scenario: Successfully listing a textbook for
   sale.
2   Given: I am logged into CampusSwap and on the
      'Sell' page.
3   When: I upload a valid image, enter the title
      'Clean Code', and set the price to '$20.'
4   And: I click 'Post Item.'
5   Then: The system should display a 'Success'
      message.
6   And: The item should appear in the 'Recently
      Added' gallery.
```


The AI Superpower: Your Simulated QA Team



Avatar 1: The User

Role: Persona-Based Discovery. Acts as a frustrated or malicious user to uncover edge cases.



Avatar 2: The Senior QA

Role: The Refinement Loop. Audits your test drafts for ambiguity and completeness.



Avatar 3: The Developer

Role: Automated Script Generation. Bridges natural language tests into Executable code.

Phase 1: Requirement Discovery (AI as User)

Act as a frustrated student trying to buy a book on a slow connection. What are 5 things that could go wrong? Write these as negative acceptance tests.

Output Excerpt

- | Negative Test 1: Image upload times out after 30 seconds.
- | Negative Test 2: Double-clicking 'Buy' due to lag charges the account twice.

The Core Lesson: Acceptance testing requires moving beyond the “Happy Path” to intentionally handle errors and edge cases.

Phase 2: The Refinement Loop (AI as Senior QA)

Prompt: I am an undergraduate SE student. Review these acceptance tests for the CampusSwap app. Identify any missing steps, vague language, or logic gaps.

...When I upload a valid image...

AI Audit Notes

Ambiguity Detected.

"Valid image" is undefined.
Specify file size and type
(e.g., < 5MB, .jpg, .png).

The Core Lesson: Precision is non-negotiable. AI forces you to define exact measurable boundaries.

The Matrix: From Flawed to Flawless

Flawed Requirement	The AI Translation Prompt	Measurable Metric
The search should be very fast.	Help me redefine 'fast' into a measurable, technical acceptance test for a search query.	The search results must populate in under 2 seconds for a query of 1,000 items.

Unquantifiable adjectives ('fast', 'secure', 'easy') are bugs in your requirements.

Phase 3: Automated Script Generation (AI as Developer)

Prompt: Convert this Gherkin scenario into a JavaScript Playwright script using Page Object Model.

When I enter the title
'Clean Code'

Gherkin Draft

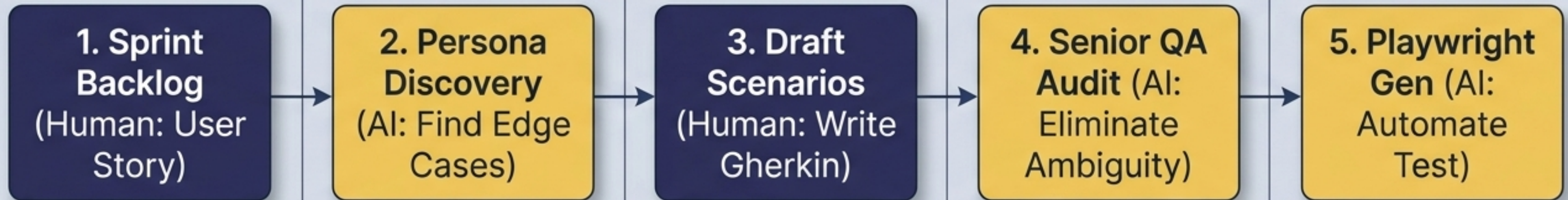


```
1 await sellPage.enterTitle('Clean Code');  
2 await expect(page.locator('.success-  
msg')).toBeVisible();
```

JavaScript (Playwright)

The Core Lesson: Modern “Low-Code” tools (Mabl, Testim) utilize this exact natural-language-to-code pipeline in professional environments.

The AI-Enhanced QA Pipeline



The modern QA Consultant doesn't just write tests—they engineer automated validation pipelines.

Measuring Mastery: QA Evaluation Rubric

✓ Foundational

Syntax Compliance

Can the student follow the Gherkin Given/When/Then syntax correctly without structural errors?

🔍 Analytical

Edge Case Coverage

Did the test suite include 'Negative Tests' (e.g., verifying system behavior when a textbook price is negative)?

💡 Advanced

AI-Assisted Discovery

Did the student utilize AI to identify a 'hidden' requirement (e.g., accessibility standards or mobile responsiveness)?



The Final Validation

**Code that works flawlessly
but solves the wrong problem
is still defective.**

Acceptance testing is the ultimate source of truth. By combining human empathy with AI precision, we ensure we are not just building software right, but building the right software.